

IIT Foundation Module for Class VI Math

- Theme 1: Number System
 1. Multiplication and division of integers
 2. Properties of operations on integers: Commutativity, associativity, existence of identity and inverse, and distributivity
 3. Problem solving using operations on integers
 4. Solution of word problems involving integers (all operations)
 5. Introduction to rational numbers (with representation on number line)
 6. Word problems on rational numbers (all operations)
 7. Decimal representation of rational numbers
 8. Problem solving using operations on rational numbers and decimal fractions
 9. Fraction as an operator
 10. Reciprocal of a fraction
 11. Multiplication and division of decimal fractions
 12. Exponents (only natural numbers)
 13. Laws of exponents (through observing patterns to arrive at generalisation)
 14. Application of laws of exponents in simple daily life problems
 15. Revision idea of sets
 16. Equal, equivalent, and universal sets
 17. Cardinal property of sets

- Theme 2: Ratio and Proportion
 1. Ratio and proportion (revision)
 2. Unitary method continued, consolidation, general expression for unitary method
 3. Percentage – an introduction
 4. Understanding percentage as a fraction with denominator 100
 5. Converting fractions and decimals into percentage and vice-versa
 6. Application to profit and loss (single transaction only)
 7. Application to simple interest (time period in complete years)
 8. Speed, distance, time

- Theme 3: Algebra

1. Terms related to algebra – constants, variables, terms, coefficient of terms, like and unlike terms, etc.
2. Generate algebraic expressions
3. Perform operations (addition and subtraction) on algebraic expressions with integral coefficients only
4. Simple linear equations in one variable (in contextual problems) with two operations
5. Inequalities and solution of simple inequalities in one variable

- Theme 4: Geometry

1. Understanding shapes
 - Pairs of angles – linear, supplementary, complementary, adjacent, vertically opposite
 - Properties of parallel lines with transversal – alternate, corresponding, interior, exterior angles
2. Properties of triangles
 - Angle sum property
 - Exterior angle property
 - Pythagoras Theorem (Verification only)
3. Symmetry
 - Recalling reflection symmetry
 - Idea of rotational symmetry, observations of rotational symmetry of 2-D objects (90° , 120° , 180°)
4. Representing 3-D in 2-D
 - Identification and counting of vertices, edges, faces, nets (for cubes, cuboids, cylinders, cones)
 - Mapping the space around approximately through visual estimation
5. Congruence
 - Congruence through superimposition
 - Extend congruence to simple geometrical shapes (e.g., triangles, circles)
 - Criteria of congruence
6. Construction
 - Construction of a line parallel to a given line from a point outside it

- Construction of simple triangles

- Theme 5: Mensuration

1. Revision of perimeter and idea of circumference of a circle
2. Area

- Theme 6: Data Handling

1. Collection and organisation of data – choosing the data to collect for hypothesis testing
2. Mean, median, and mode of ungrouped data – understanding what they represent
3. Constructing and interpreting bar graphs
4. Feel of probability using data through experiments – notion of chance in events like tossing coins, dice, etc., tabulating and counting occurrences of 1 through 6 in a number of throws, comparing the observation with that for a coin, observing strings of throws, notion of randomness